

ESTIMATIONS OF POPULATION PARAMETERS IN THE PRESENCE OF CENSORED AND TRUNCATED SAMPLES

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Summary: Censored and truncated data are becoming more common in environmental, toxicological, and biomedical studies. This paper is concerned with the problem of estimating the mean and standard deviation of normal populations from truncated, and from both censored and truncated samples. Maximum likelihood estimating equations are derived which can easily be solved by simple iterative processes. Easily computed estimates of the population mean and standard deviation are obtained from (1) truncated normally distributed samples, (2) left truncated and left censored normally distributed samples, and (3) left truncated and right censored normally distributed samples. A new algorithm for obtaining the maximum likelihood estimates of μ and σ from singly-truncated samples is provided which does not require auxiliary tables. The algorithm utilizes the *R* language. Worked examples are given illustrating the use of these estimators utilizing computer programs whose codes are given in the Appendix.

1. Introduction

In life practising, toxicological studies, biological assays, target analysis, dosage-response studies, and in many other estimation problems, sample observations are often restricted in some portions of the range of possible

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population values. In environmental studies, scientists identify and measure concentration of environmental pollutants, and establish safe risk levels. When concentrations of environmental pollutants fall below these risk levels, no further action is taken. However, when concentrations fall above these risk levels samples are collected. In statistical terminology, these risk levels are called truncated limits (TL) and data collected above these levels are said to be left truncated. A sample is said to be a left truncated if only concentrations greater than or equal to a known truncation limit TL are used. The number of concentrations falling below TL and their actual values are ignored or unknown. A sample is said to be right truncated if only concentrations less than or equal to a known truncation limit TL are used. The number of concentrations falling above TL and their actual values are ignored or unknown. A sample for which some observations are known only to fall below a known detection limit, while the remaining observations falling above the detection limit are fully measured and reported is called left-singly censored. Left-truncated data sets with a truncation limit TL that contain left-censored observations with a detection limit (DL), ($DL > TL$), are called left-truncated and left-censored. Such data commonly arise in environmental and toxicological studies. Left-truncated data sets with a truncation limit TL that contain right-censored observations with a detection limit (DL), ($DL > TL$), are called left-truncated and right-censored. Such data commonly arise in biomedical studies. The lognormal is the most widely used probability distribution to fit to these data because of the observed right-skewness of many environmental data Gilbert (1987). However, Shumway et al. (1989) examined transformations to normality from among the Box and Cox (1964) family of transformations:

$Y = \frac{X^\lambda - 1}{\lambda}$ for $\lambda \neq 0$, and $Y = \ln(X)$ for $\lambda = 0$. The transformed variable Y is assumed to be normally distributed with mean μ and standard deviation σ . Since the logarithm of a log-normally distributed random variable is normally distributed, the methods introduced in this article can be used to estimate parameters of a lognormal population.

Cohen (1959) used the method of maximum likelihood to derive estimators for the μ and σ parameters from truncated and from censored samples. Dempster et al. (1977) proposed an iterative method, called the Expectation Maximization algorithm, for obtaining the maximum likelihood estimates from normally distributed censored samples. Other suggested methods for estimating population parameters from censored samples are discussed in Aboueissa and Stoline (2004, 2006), Aboueissa (2008), Gibbons (1994), Gleit (1985), Schneider (1986), Gupta (1952), Stoline (1993), El-Shaarawi and Dolan (1989), El-Shaarawi and Esterby (1992), USEPA (1989), NCASI (1985, 1991), Gilliom and Helsel (1986), Helsel and Gilliom (1986), Helsel and Hirsch (1988), Schmee et al. (1985), Hald (1949), Hass and Scheff (1990), Hyde (1977), Kushner (1976), John Hyde (1977), Lagakos et al. (1988), Tsai (1990), and Wolynetz (1979).

This paper has four goals. The first goal is to develop a new algorithm for computing Cohen's (1959) maximum likelihood estimates of population parameters from truncated samples. Cohen (1959) introduced maximum likelihood estimators for population parameters from truncated samples. Cohen (1959) also provided tables that are needed to report these estimates. However, interpolation is often required in these tables. In addition, these tables are restricted for values of the auxiliary parameter γ_T used in these tables. The new algorithm for computing Cohen's maximum likelihood

estimates from truncated samples provides more adequate estimates for the population parameters and handles more situations than Cohen's (1959) estimators do, since tables used to calculate the Cohen γ_T -parameter require interpolation. These estimators reduce the computation effort to determine the maximum likelihood estimates substantially by solving a nonlinear equation for the argument that is required to calculate the γ_T -parameter. The maximum likelihood estimators obtained via this method require no tables and are more easily computed than the maximum likelihood estimators of Cohen (1959). Cohen's (1959) estimators cannot be used to obtain the maximum likelihood estimates from truncated samples that yield $\gamma_T < 0.06246$ or $\gamma_T > 0.880$, but the new algorithm can be used to calculate these estimates for any value of γ_T . This algorithm can be employed to construct a more extensive tabulation of the γ_T -parameter for values more than the ones given by Cohen (1959). The second goal is to develop easily computed maximum likelihood estimates of population parameters obtained from samples which are left truncated and left censored. The third goal is to develop easily computed maximum likelihood estimates of population parameters obtained from samples which are left truncated and right censored. Estimators of population means and variances derived in this article are expressed as restricted sample means and variances respectively, plus simple easily computed corrections which involve only a single auxiliary function of the sample terminus. The fourth goal is to provide R programs for computing maximum likelihood estimates from the samples under consideration since there is no explicit software available to calculate these estimates of population parameters from samples mentioned above. These programs are very fast and efficient. It is easy to use and modify for the relevant use. Worked examples are given illustrating the use of these

estimators utilizing these R programs whose codes are given in the Appendix section.

2. Assumptions and Notations

Consider a random sample $x_1, \dots, x_m, x_{m+1}, \dots, x_n$ of size n , taken from a normal population with mean μ and standard deviation σ . Only the m observations $x_1, \dots, x_m$ are non-censored and the $m_c = n - m$ observations $x_{m+1}, \dots, x_n$ are left censored. For censored observations, it is only known that $x_{m+i} < DL$ for $i = 1, \dots, m_c$. Let the functions $\phi(\xi)$ and $\Phi(\xi)$ be the *pdf* and *cdf* of the standard unit normal. For the above sample we define:

$$\bar{x}_m = \frac{1}{m} \sum_{i=1}^m x_i \quad \text{and} \quad s_m^2 = \frac{1}{m} \sum_{i=1}^m (x_i - \bar{x}_m)^2 \quad (2.1)$$

which are the sample mean and sample variance of the m non-censored observations $x_1, \dots, x_m$. For truncation data, let $x_1, x_2, \dots, x_n$ be a random sample of size n , where $x_i > TL$ for $i = 1, 2, \dots, n$, and TL is a known value. For this type of truncated samples we define:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad \text{and} \quad s^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \quad \text{and} \quad \gamma_T = \frac{s^2}{(\bar{x} - TL)^2} \quad (2.2)$$

We also define

$$\begin{aligned} \Phi(\xi_D) &= \int_{-\infty}^{\xi_D} \phi(t) dt, \quad Z(\xi_D) = \frac{\phi(\xi_D)}{[1 - \Phi(\xi_D)]}, \quad h = \frac{m_c}{m}, \quad h_m = \frac{m}{n} \\ \text{and } \Phi(\xi_T) &= \int_{-\infty}^{\xi_T} \phi(t) dt, \quad Z(\xi_T) = \frac{\phi(\xi_T)}{[1 - \Phi(\xi_T)]}, \\ \text{where } \xi_D &= \frac{DL - \mu}{\sigma} \quad \text{and} \quad \xi_T = \frac{TL - \mu}{\sigma} \end{aligned} \quad (2.3)$$

It is convenient to define the following, using the notation in (2.1)–(2.3)

$$\bar{\xi}_{TD} = \frac{\xi_T + \xi_D}{2} \text{ and } \gamma_{TD} = \frac{s_m^2}{(\bar{x}_m - \overline{TDL})^2}, \text{ where } \overline{TDL} = \frac{TL + DL}{2}. \quad (2.4)$$

It should be noted that the probability density function, $\phi(\cdot)$, of $N(\mu, \sigma^2)$ is symmetrical about μ , therefore truncation of $\phi(-x)$ on the left at $-TL$ is equivalent to truncation of $\phi(x)$ on the right at TL . Thus the maximum likelihood estimators obtained from truncated samples in Section 3 are equally applicable to both singly-left and singly-right truncated samples. It should also be observed that using a transformation technique makes non-normally distributed random variables normally distributed. For example, the logarithm of a log-normally distributed random variable is normally distributed. Thus estimators described in this article can be used to estimate population parameters from non-normally distributed samples.

Outline of the Paper:

A new computer algorithm for obtaining the Cohen (1959) maximum likelihood estimates of μ and σ from samples that are singly truncated is introduced in Section 3. The maximum likelihood estimates obtained via this method are essentially Cohen's (1959) estimates, which are obtained without the use of any auxiliary tables. The algorithm utilizes the R language. These estimates are examined and compared with those introduced by Cohen (1959). The method of maximum likelihood is employed to obtain maximum likelihood estimators for the mean and standard deviation from normally distributed left-truncated and left- (or right-) censored samples with $DL > TL$ in Sections 4 and 5. Illustrative examples are given illustrating the use of these estimators of obtaining estimates of population parameters from left-truncated data, and from both left-truncated and left- (or right)-censored data sets in Section 6. Computer programs for computing Cohen's λ -parameter as

a function of γ_T , and for computing maximum likelihood estimates from the samples under consideration, are written in the R language whose codes are provided in the Appendix Section. These programs are very fast, efficient, and it is easy to use and modify for the relevant use. Conclusions and recommendations are given in Section 7.

3. A new algorithm for calculating Cohen's Maximum Likelihood Estimators of μ and σ from truncated samples

Let $x_1, x_2, \dots, x_n$ be a random sample of size n truncated on the left, where $x_i \geq TL$ for $i = 1, 2, \dots, n$. Let TL be the left truncation limit, which is a known fixed value. The likelihood function for a normally distributed singly-left truncated sample can be written as

$$L(\mu, \sigma) = \frac{1}{[1 - \Phi(\xi_T)]^n} \frac{1}{[\sigma\sqrt{\pi}]^n} e^{-\frac{1}{2(\sigma)^2} \sum_{i=1}^n (x_i - \mu)^2} \quad (3.1)$$

Cohen (1959) employed the method of maximum likelihood to obtain the maximum likelihood estimators for the mean and standard deviation from normally distributed singly-truncated samples and developed the system of equations:

$$\bar{x} - \mu = \sigma Z(\xi_T), \quad (3.2)$$

$$s^2 + (\bar{x} - \mu)^2 = \sigma^2 [1 + \xi_T Z(\xi_T)], \quad (3.3)$$

and

$$TL - \mu = \sigma \xi_T \quad (3.4)$$

Solving this system of equations (3.2)–(3.4) for μ and σ leads to the following system of estimating equations:

$$\mu_T = \bar{x} - \lambda_T (\bar{x} - TL), \quad (3.5)$$

$$\sigma_T = \sqrt{s^2 + \lambda_T (\bar{x} - TL)^2}, \quad (3.6)$$

and

$$\gamma_T = \frac{[1 - Z(\xi_T)(Z(\xi_T) - \xi_T)]}{[Z(\xi_T) - \xi_T]^2} \quad (3.7)$$

where

$$\lambda_T = \lambda_T(\xi_T) = \frac{Z(\xi_T)}{[Z(\xi_T) - \xi_T]} \quad (3.8)$$

Cohen's (1959) method is based on estimating the λ -parameter λ_T as a function of $\gamma_T = \frac{s^2}{(\bar{x} - TL)^2} = \frac{[1 - Z(\xi_T)(Z(\xi_T) - \xi_T)]}{[Z(\xi_T) - \xi_T]^2}$. Cohen (1959) provided tables of the function λ_T in terms of γ_T . However, interpolation is often required in these tables, and for this reason the method may not be adequate for routine use. In addition, these tables are restricted for values of $\gamma_T \in [0.06246, 0.880]$.

The new algorithm to approximate the solution to the maximum likelihood estimating equations by use of a computer program is based on estimating the λ -parameter $\lambda_T = \lambda_T(\xi_T) = \frac{Z(\xi_T)}{[Z(\xi_T) - \xi_T]}$ as a function not of γ_T , but of ξ_T , which requires solving the rather complex non-linear estimating equation (3.7) for ξ_T . Because of the difficulty of solving equation (3.7) explicitly for ξ_T , an iterative method should be used. Equation (3.7) can be written as

$$g(\xi_T) = \gamma_T \xi_T^2 + (\gamma_T + 1) Z^2(\xi_T) - (2\gamma_T + 1) Z(\xi_T) \xi_T - 1 = 0 \quad (3.9)$$

There is no explicit solution for the equation $g(\xi_T) = 0$ because of its complexity. However, more investigation done by the author, using both numerical calculations and graphical techniques, shows that it has a unique real solution as shown in Figure 1. This solution can be obtained via a computer program whose code is given in the Appendix. Since there is no closed-form

solution, a grid search method is used in an algorithm to obtain the solution of $g(\xi_T) = 0$, $\hat{\xi}_T$, given $\gamma_T = \frac{s_2}{(\bar{x} - TL)^2}$.

More details are now given in the algorithm yielding the MLEs for μ and σ utilizing a grid search. Let $\underline{\xi}_T$ be a vector of the values in the sequence: $-5.0000, -4.9999, -4.9998, \dots, 4.9998, 4.9999, 5.0000$. For convenience, let this vector sequence be denoted as: $\underline{\xi}_T = seq(-5, 5, 0.00001)$. Let $\underline{\xi}_T(i)$ be the i^{th} element of $\underline{\xi}_T$, where $\underline{\xi}_T(1) = -5.0000$. With γ_T obtained from sample data, let $\underline{g}$ be a vector with the element $\underline{g}(i) = \underline{g}(\underline{\xi}_T(i))$, where $\underline{g}(1) = \underline{g}(\underline{\xi}_T(1)) = \underline{g}(-5.0000)$. The following criteria are used to find the solution (root) of $g(\xi_T) = 0$, by using linear interpolation.

$$\text{If } (\underline{g}(i) < 0 \quad \text{and} \quad \underline{g}(i+1) > 0),$$

the solution of $g(\xi_T) = 0$ is the interpolated point

$$\hat{\xi}_T = \left(\frac{\underline{\xi}_T(i+1) - \underline{\xi}_T(i)}{\underline{g}(\underline{\xi}_T(i+1)) - \underline{g}(\underline{\xi}_T(i))} \right) \underline{g}(\underline{\xi}_T(i)) + \underline{\xi}_T(i) \quad (3.10)$$

With $\hat{\xi}_T$ obtained from (3.10), an estimate of the λ -parameter $\lambda_T = \lambda_T(\xi_T)$. From (3.8) is computed as $\hat{\lambda}_T = \lambda_T(\hat{\xi}_T) = \frac{Z(\hat{\xi}_T)}{[Z(\hat{\xi}_T) - \hat{\xi}_T]}$, and consequently the MLEs of μ and σ obtained from (3.5) and (3.6) as

$$\hat{\mu}_T = \bar{x} - \hat{\lambda}_T (\bar{x} - TL), \quad (3.11)$$

and

$$\hat{\sigma}_T = \sqrt{s^2 + \hat{\lambda}_T (\bar{x} - TL)^2}, \quad (3.12)$$

where

$$\hat{\lambda}_T = \lambda_T(\hat{\xi}_T) = \frac{Z(\hat{\xi}_T)}{[Z(\hat{\xi}_T) - \hat{\xi}_T]}, \quad (3.13)$$

and $\hat{\xi}_T$ is the numerical solution of (3.7).

It should be noted that the maximum likelihood estimators given by (3.11)–(3.12), obtained from singly-left truncated samples, are applicable to singly-right truncated samples as well since the truncation of $\phi(-x)$ on the left at $-TL$ is equivalent to truncation of $\phi(x)$ on the right at TL .

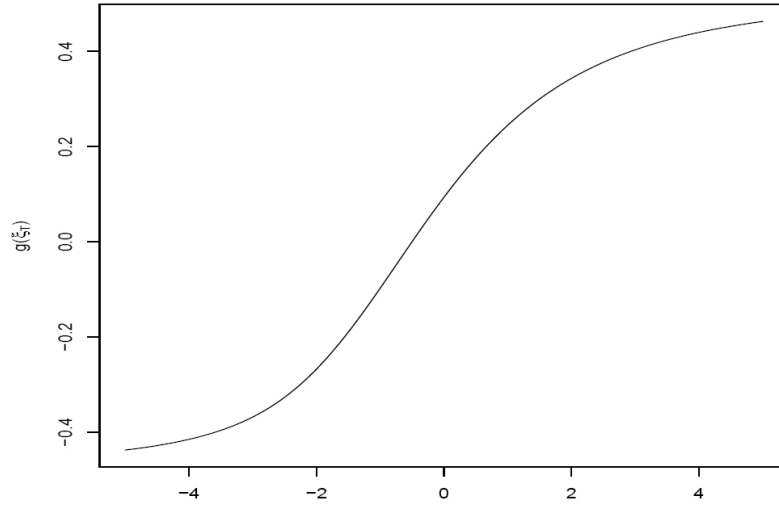


Figure 1 The typical plot of $g(\xi_T)$ given γ_T

These estimators will be referred to as MLELRT. Based on the new algorithm, a computer program for obtaining the tabulated Cohen λ -parameter, λ_T , as a function of γ_T is introduced. The program is called “Abou.Cohen.lambdat.hat”. Copy of the source code is given in the Appendix. The new procedure used in the new algorithm is more accurate than the procedure described by Cohen (1959) since tables used to calculate the Cohen λ -parameter require interpolation. It should be noted that the new algorithm can be employed to construct a more extensive tabulation of the λ -parameter values than the ones given by Cohen (1959).

4. Maximum likelihood estimators of μ and σ from both left-truncated and left-censored samples

Maximum likelihood estimators of parameters of the normal distribution are derived for samples that are singly left truncated and left censored. Let $x_1, x_2, \dots, x_n$ be a random sample of size n truncated on the left, where $x_i \geq TL$ for $i = 1, 2, \dots, n$, such that m observations $x_1, \dots, x_m$ are non-censored and the $m_c = n - m$ observations $x_{m+1}, \dots, x_n$ are left censored. For left-censored observations, it is only known that $x_{m+i} < DL$ for $i = 1, \dots, m_c$. Let TL and DL be the truncation limit and the detection limit, which are known fixed values. The likelihood function for the sample under consideration can be written as

$$L(\mu, \sigma) = \frac{1}{[1 - \Phi(\xi_T)]^n} \frac{n!}{m_c! m!} [\Phi(\xi_D)]^{m_c} \left[\frac{1}{\sigma \sqrt{2\pi}} \right]^m e^{-\frac{1}{2\sigma^2} \sum_{i=1}^m (x_i - \mu)^2} \quad (4.1)$$

Hence, the corresponding log-likelihood function of (4.1) is given by

$$\begin{aligned} \ell(\mu, \sigma) = & -n \ln [1 - \Phi(\xi_T)] + \ln \left[\frac{n!}{m_c! m!} \right] + m_c \ln [\Phi(\xi_D)] - m \ln (\sqrt{2\pi}) \\ & - m \ln (\sigma) - \frac{1}{2\sigma^2} \sum_{i=1}^m (x_i - \mu)^2 \end{aligned} \quad (4.2)$$

Equating the partial derivatives of $\ell(\mu, \sigma)$ with respect to μ and σ to zero yields

$$\bar{x}_m - \mu = \sigma \left[\frac{1}{h_m} Z(\xi_T) + h Z(-\xi_D) \right] \quad (4.3)$$

and

$$\sigma^2 = s_m^2 + (\bar{x}_m - \mu)^2 - \sigma^2 \left[\frac{1}{h_m} Z(\xi_T) \xi_T + h Z(-\xi_D) (\xi_D) \right] \quad (4.4)$$

From (2.3) we have

$$TL - \mu = \sigma \xi_T \quad (4.5)$$

and

$$DL - \mu = \sigma \xi_D \quad (4.6)$$

Eliminating $(\bar{x}_m - \mu)^2$ between (4.3) and (4.4) leads to

$$\sigma^2 = s_m^2 + \sigma^2 Y(\xi_T) + \sigma^2 Y_L(\xi_D) + \sigma^2 \left[2 \frac{h}{h_m} Z(\xi_T) Z(-\xi_D) \right] \quad (4.7)$$

where

$$Y(\xi_T) = \frac{Z(\xi_T)}{h_m} \left[\frac{Z(\xi_T)}{h_m} - \xi_T \right] \text{ and } Y_L(\xi_D) = h Z(-\xi_D) [h Z(-\xi_D) - \xi_D].$$

Combining (4.5) and (4.6) yields

$$\mu = \overline{TDL} - \sigma \bar{\xi}_{TD} \quad (4.8)$$

Eliminating μ between (4.3) and (4.8) yields

$$\sigma^2 = \frac{(\bar{x}_m - \overline{TDL})^2}{Y_L^2(\xi_T, \xi_D)} \quad (4.9)$$

where $Y_L(\xi_T, \xi_D) = \left[\frac{1}{h_m} Z(\xi_T) + h Z(-\xi_D) \right] - \bar{\xi}_{TD}$.

Combining (4.7) and (4.9) with replacing σ by σ_{LTLC} yields

$$\sigma_{LTLC} = \sqrt{s_m^2 + \lambda_{\sigma_{LTLC}} (\bar{x}_m - \overline{TDL})^2} \quad (4.10)$$

where

$$\lambda_{\sigma_{LTLC}} = \frac{1}{Y_L^2(\xi_T, \xi_D)} \left[Y(\xi_T) + Y_L(\xi_D) + 2 \frac{h}{h_m} Z(\xi_T) Z(-\xi_D) \right] \quad (4.11)$$

Solving (4.3) for μ with replacing μ by μ_{LTLC} leads to

$$\mu_{LTLC} = \bar{x}_m - \lambda_{\mu_{LTLC}} \sigma_{LTLC} \quad (4.12)$$

where

$$\lambda_{\mu_{LTLC}} = \frac{1}{h_m} Z(\xi_T) + h Z(-\xi_D) \quad (4.13)$$

Eliminating σ^2 between (4.7) and (4.9) yields

$$\gamma_{LTLC} = \frac{1}{Y_L^2(\xi_T, \xi_D)} [1 - Y(\xi_T) - Y_L(\xi_D) - 2 \frac{h}{h_m} Z(\xi_T) Z(-\xi_D)] \quad (4.14)$$

To obtain the desired MLEs of μ and σ from (4.10)–(4.14), it is necessary to solve the rather complex non-linear estimating equation (4.14) for estimates $\hat{\xi}_T$ and $\hat{\xi}_D$. Because of the difficulty of solving this equation explicitly for ξ_T and ξ_D , an iterative method should be used. The iterative method described here is based on reducing the unknown parameters in (4.14) to only one parameter (say ξ_T) using the relationship between ξ_T and ξ_D .

It is recognized that

$$\xi_D = \xi_T + d_{LTLC} \quad \text{for some } d_{LTLC} > 0. \quad (4.15)$$

Since for any sample the parameter values n , m_c are known, and TL , $\bar{x}_m$ and s_m^2 can be obtained from the sample data, then the following relations will usually provide satisfactory first approximations ξ_{D_f} to ξ_D and ξ_{T_f} to ξ_T :

$$h_{m_c} = \frac{m_c}{n} = \int_{-\infty}^{\xi_D} \phi(t) dt = \Phi(\xi_D) \quad (4.16)$$

with

$$\xi_{D_f} = \Phi^{-1}(h_{m_c}) \quad (4.17)$$

and

$$\xi_{T_f} = \frac{TL - \bar{x}_m}{s_m} \quad (4.18)$$

An estimate of d_{LTLC} is obtained from (4.15) as

$$\hat{d}_{LTLC} = \xi_{D_f} - \xi_{T_f}. \quad (4.19)$$

Using expression (4.15) with $\hat{d}_{LTLC}$ obtained from (4.19), the estimating equation (4.14) can be written as

$$g(\xi_T, \hat{d}_{LTLC}) = \frac{1}{Y_L^2(\xi_T, \hat{d}_{LTLC})} - \lambda_{\sigma_{LTLC}}(\xi_T, \hat{d}_{LTLC}) - \gamma_{LTLC} \quad (4.20)$$

which behaves like a second-degree function of ξ_T . There is no explicit solution for the equation $g(\xi_T, \hat{d}_{LTLC}) = 0$ because of its complexity. The typical plots of $g(\xi_T, \hat{d}_{LTLC})$ as a function of ξ_T are given in Figure 2. More investigation done by the author, using both numerical calculations and graphical techniques, shows that for some data sets (4.20) has a unique solution, say $\hat{\xi}_T$, and for some other data sets it has two distinct real solutions as shown in Figure 2. Let us call these two roots the minimum and the maximum roots. Since there is no closed-form solution, a grid search method is used in an algorithm to obtain the solution of $g(\xi_T, \hat{d}_{LTLC}) = 0$. In case of $g(\xi_T, \hat{d}_{LTLC}) = 0$ having two solutions, further investigation has shown that the minimum solution, say $\hat{\xi}_T$, leads to the desired maximum likelihood estimators.

More details are now given in the algorithm yielding the MLE estimators for μ and σ utilizing a grid search. Let $\underline{\xi}_T$ be a vector of the values in the sequence: $-5.0000, -4.9999, -4.9998, \dots, 4.9998, 4.9999, 5.0000$. For convenience let this vector sequence be denoted as: $\underline{\xi}_T = seq(-5, 5, 0.00001)$. Let $\underline{\xi}_T(i)$ be the i^{th} element of $\underline{\xi}_T$, where $\underline{\xi}_T(1) = -5.0000$. With γ_T , h_{mc} , h , TL and DL obtained from sample data, let $\underline{g}$ be a vector with the i^{th} element $\underline{g}(i) = g(\underline{\xi}_T(i))$, where $\underline{g}(1) = g(\underline{\xi}_T(1)) = g(-5.0000)$. The following criteria are used to find the solution $\hat{\xi}_T$ of $g(\xi_T, \hat{d}_{LTLC}) = 0$, by using linear interpolation.

In case of $g(\xi_T, \hat{d}_{LTLC}) = 0$ having two solutions:

$$\text{If } (\underline{g}(i) > 0 \quad \text{and} \quad \underline{g}(i+1) < 0)$$

and in case of $g(\xi_T, \hat{d}_{LTLC}) = 0$ having one solution:

$$\text{If } (\underline{g}(i) < 0 \quad \text{and} \quad \underline{g}(i+1) > 0)$$

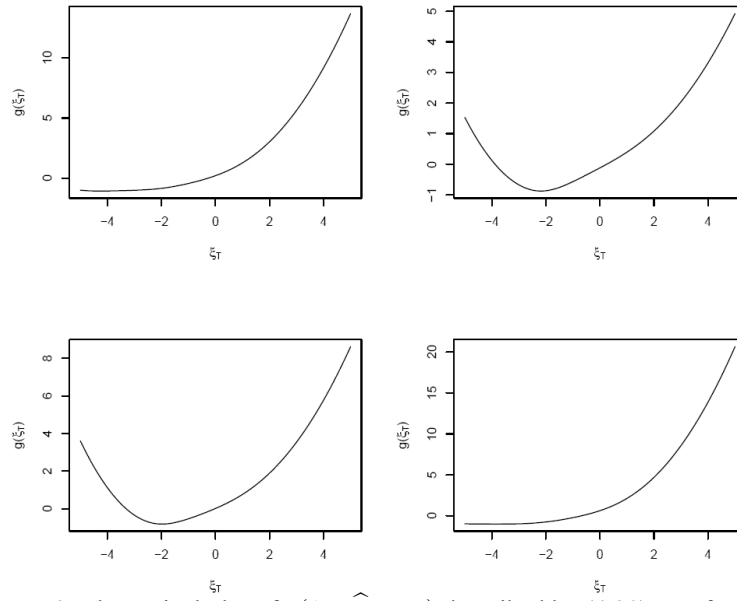


Figure 2 The typical plot of $g(\xi_T, \hat{d}_{LTLC})$ described by (4.20) as a function of ξ_T

then $\hat{\xi}_T$ is given by

$$\hat{\xi}_T = \left(\frac{\xi_T(i+1) - \xi_T(i)}{\underline{g}(\xi_T(i+1)) - \underline{g}(\xi_T(i))} \right) \underline{g}(\xi_T(i)) + \underline{\xi}_T(i) \quad (4.21)$$

With $\hat{\xi}_T$ obtained from (4.21) and $\hat{d}_{LTLC}$ obtained from (4.19), an estimate of ξ_D follows from (4.15) as

$$\hat{\xi}_D = \hat{\xi}_T + \hat{d}_{LTLC} \quad (4.22)$$

With $\hat{\xi}_T$ and $\hat{\xi}_D$ determined by (4.21)-(4.22), the MLE estimators for μ and σ follow from (4.10)–(4.13) as

$$\hat{\sigma}_{LTLC} = \sqrt{s_m^2 + \hat{\lambda}_{\sigma_{LTLC}} (\bar{x}_m - \overline{TDL})^2} \quad (4.23)$$

$$\hat{\mu}_{LTLC} = \bar{x}_m - \hat{\lambda}_{\mu_{LTLC}} \hat{\sigma}_{LC} \quad (4.24)$$

where

$$\hat{\lambda}_{\sigma_{LTLC}} = \frac{1}{Y_L^2(\hat{\xi}_T, \hat{\xi}_D)} \left[Y(\hat{\xi}_T) + Y_L(\hat{\xi}_D) + 2 \frac{h}{h_m} Z(\hat{\xi}_T) Z(-\hat{\xi}_D) \right] \quad (4.25)$$

and

$$\hat{\lambda}_{\mu_{LTLC}} = \frac{1}{h_m} Z(\hat{\xi}_T) + h Z(-\hat{\xi}_D) \quad (4.26)$$

These estimators will be referred to as MLELTLC.

5. Maximum likelihood estimators of μ and σ from both left-truncated and right-censored samples

In this section the method of maximum likelihood is employed to derive estimators of population parameters from left-truncated and right-censored data. Assume that there exists a random sample of n data values: $x_1, x_2, \dots, x_n$ truncated on the left, where $x_i \geq TL$ for $i = 1, 2, \dots, n$, taken from a normal population with mean μ and standard deviation σ . For convenience, let us assume that the first m observations $x_1, \dots, x_m$ are non-censored (fully measured) and the remaining $m_c = n - m$ observations $x_{m+1}, \dots, x_n$ are right-censored. For right-censored observations, it is only known that $x_{m+i} > DL$ for $i = 1, \dots, m_c$. Let TL and DL be the truncation and the detection limits, which are known fixed values. The likelihood function for the sample under consideration is given by

$$L(\mu, \sigma) = \frac{1}{[1 - \Phi(\xi_T)]^n} \frac{n!}{m_c! m!} [1 - \Phi(\xi_D)]^{m_c} \left[\frac{1}{\sigma \sqrt{2\pi}} \right]^m e^{-\frac{1}{2\sigma^2} \sum_{i=1}^m (x_i - \mu)^2} \quad (5.1)$$

Hence, the corresponding log-likelihood function of (4.1) is given by

$$\begin{aligned} \ell(\mu, \sigma) = & -n \ln [1 - \Phi(\xi_T)] + \ln \left[\frac{n!}{m_c! m!} \right] + m_c \ln [1 - \Phi(\xi_D)] - m \ln (\sqrt{2\pi}) \\ & - m \ln(\sigma) - \frac{1}{2\sigma^2} \sum_{i=1}^m (x_i - \mu)^2 \end{aligned} \quad (5.2)$$

By differentiating equation (5.2) with respect to μ and σ , and then equating the resulting two equations to zero, lead to the following maximum likelihood estimating equations:

$$\bar{x}_m - \mu = \sigma \left[\frac{1}{h_m} Z(\xi_T) - h Z(\xi_D) \right] \quad (5.3)$$

and

$$\sigma^2 = s_m^2 + (\bar{x}_m - \mu)^2 - \sigma^2 \left[\frac{1}{h_m} Z(\xi_T) \xi_T - h Z(\xi_D) \xi_D \right], \quad (5.4)$$

where $Z(\xi_D) = \frac{\phi(\xi_D)}{1 - \Phi(\xi_D)}$.

With $\xi_{D_f} = \Phi^{-1}(1 - h_{m_c})$ the maximum likelihood estimators, $\hat{\mu}_{LTRC}$ and $\hat{\sigma}_{LTRC}$, for μ and σ are then given by

$$\hat{\sigma}_{LTRC} = \sqrt{s_m^2 + \hat{\lambda}_{\sigma_{LTRC}} (\bar{x}_m - \overline{TDL})^2}, \quad (5.5)$$

and

$$\hat{\mu}_{LTRC} = \bar{x}_m + \hat{\lambda}_{\mu_{LTRC}} \hat{\sigma}_{LTRC}, \quad (5.6)$$

with

$$\hat{\lambda}_{\sigma_{LTRC}} = \frac{1}{Y_R^2(\hat{\xi}_T, \hat{\xi}_D)} \left[Y(\hat{\xi}_T) + Y_R(\hat{\xi}_D) - 2 \frac{h}{h_m} Z(\hat{\xi}_T) Z(\hat{\xi}_D) \right], \quad (5.7)$$

and

$$\hat{\lambda}_{\mu_{LTRC}} = \frac{1}{h_m} Z(\hat{\xi}_T) - h Z(\hat{\xi}_D), \quad (5.8)$$

where $\hat{\xi}_T$ and $\hat{\xi}_D$ are the solution of

$$\gamma_{LTRC} = \frac{1}{Y_R^2(\xi_T, \xi_D)} \left[1 - Y(\xi_T) - Y_R(\xi_D) + 2 \frac{h}{h_m} Z(\xi_T) Z(\xi_D) \right], \quad (5.9)$$

where $Y_R(\xi_D) = h Z(\xi_D) [h Z(\xi_D) + \xi_D]$. It should be noted that $\hat{\xi}_T$ and $\hat{\xi}_D$ can be obtained via an iterative method as shown in Section 4.

These estimators will be referred to as MLELTC.

6. Computer programs

To facilitate the application of these MLE estimators of μ and σ from both singlyleft truncated, and singly-left truncated and censored samples developed in this paper, computer programs were written in the R language. For the singly-left truncated samples, the program is called “MLELRT.Single.Truncated”, for the singly-left truncated and singly-left censored samples, the program is called “MLELTLC.Singly.LeftTrunc.LeftCens”, and for the singly-left truncated and singly-right censored samples, the program is called “MLELTC.Singly.LeftTrunc.RightCens”. In addition, a computer program for obtaining the Cohen λ -parameter as a function of γ_T is introduced. This program is called “Abou.Cohen.lambdat.hat”. Copies of the source codes are given in the Appendix.

7. Worked examples

Example 1 Singly-Left Truncated Data

A simulated data set of size $n = 20$ was obtained from a truncated normal population with mean $\mu = 15$ and standard deviation $\sigma = 5$. The truncation limit is assumed to be the 20th percentile of a normal population with mean $\mu = 15$ and standard deviation $\sigma = 5$. The left-truncated data are:

18.91943	19.96156	19.13235	16.75358	18.64818	12.42409	21.02902
11.14165	19.44774	15.14847	22.42261	14.62704	13.56455	16.07526
18.82385	17.95926	12.40669	10.84799	14.77135	12.88172	

The sample mean and the sample standard deviation of the these data are $\bar{x} = 16.34932$ and $s = 3.5437$, respectively. For this data set $\gamma_T = \frac{s^2}{(\bar{x} - TL)^2} = \frac{11.25179}{(16.34932 - 10.79189)^2} = 0.36431$. MLE's of μ and σ are now computed using both MLELRT estimators (3.11)–(3.12) and MLE's described in Cohen (1959).

Cohen's MLE's: By linear interpolation in Table 1 Cohen (1959), we obtain $\hat{\lambda}_T = 0.19318$ (in Cohen 1959 denoted as $\hat{\theta}$). Accordingly, the maximum likelihood estimates of μ and σ are given by:

$$\hat{\mu} = 16.34932 - 0.19318(16.34932 - 10.79189) = 15.27574$$

and

$$\hat{\sigma} = \sqrt{11.25179 + 0.19318(16.34932 - 10.79189)^2} = 4.14948$$

MLELRT estimators: The maximum likelihood estimates obtained via these estimators are obtained by applying the computer program “*MLELRT.Single.Truncated*” for the above data set as shown in the Appendix. Accordingly, we obtain

$$\hat{\xi}_T = -1.08054 \quad \text{and} \quad \hat{\lambda}_T = 0.19320$$

$$\hat{\mu}_T = 15.27568$$

and

$$\hat{\sigma}_T = 4.14951$$

Inspection of the previous estimates reveals that both methods yield quite similar results for the estimations of μ and σ .

Example 2. Singly-Right Truncated Data (This example is taken from Cohen 1959)

To insure meeting a maximum weight specification of 12.00 ounces on a certain radio component for an aircraft installation, all production of this component is weighed and those units which exceed the specified maximum weight are discarded. For a random sample of 50 units selected from the screened (accepted) production, $\bar{x} = 9.35$ ounces, $s^2 = 1.1264$, and $TL = 12.00$. Accordingly, $\gamma_T = \frac{s^2}{(\bar{x} - TL)^2} = \frac{1.1264}{(9.35 - 12.00)^2} = 0.16040$.

Cohen's MLE's: By linear interpolation in Table 1 Cohen (1959), we obtain $\hat{\lambda}_T = 0.00916$ (in Cohen 1959 denoted as $\hat{\theta}$). Accordingly, the maximum likelihood estimates of μ and σ given by:

$$\hat{\mu} = 9.35 - 0.00916(9.35 - 12.00) = 9.37427$$

and

$$\hat{\sigma} = \sqrt{1.1264 + 0.00916(9.35 - 12.00)^2} = 1.09120$$

MLELRT estimators: The maximum likelihood estimates obtained via these estimators are obtained by applying the computer program “*MLELRT.Single.Truncated*” with $\bar{x} = 9.35$, $s^2 = 1.1264$, and $TL = 12.00$ as shown in the Appendix. Accordingly, we obtain

$$\hat{\xi}_T = -2.40620 \quad \text{and} \quad \hat{\lambda}_T = 0.00916$$

$$\hat{\mu}_T = 9.37427$$

and

$$\hat{\sigma}_T = 1.09120$$

Once again, inspection of the previous estimates reveals that both methods yield same results for the estimations of μ and σ .

Example 3

To insure meeting a maximum weight specification of 12.00 ounces on a certain radio component for an aircraft installation, all production of this component is weighed and those units which exceed the specified maximum weight are discarded. For a random sample of 50 units selected from the screened (accepted) production, $\bar{x} = 7.15$ ounces, $s^2 = 1.1264$, and $TL = 12.00$. Accordingly, $\gamma_T = \frac{s^2}{(\bar{x} - TL)^2} = \frac{1.1264}{(7.15 - 12)^2} = 0.04789$.

Cohen's MLE's: With $\gamma_T = 0.04789$ the maximum likelihood estimators of μ and σ cannot be computed by Cohen's method since Cohen's tables of the function λ_T are restricted for values of $\gamma_T \in [0.06246, 0.880]$.

MLELRT estimators: The maximum likelihood estimates obtained via these estimators are obtained by applying the computer program "*MLELRT.Single.Truncated*" with $\bar{x} = 7.15$, $s^2 = 1.1264$, and $TL = 12$ as shown in the Appendix. Accordingly, we obtain

$$\hat{\xi}_T = -4.569548 \quad \text{and} \quad \hat{\lambda}_T = 0.00000255$$

$$\hat{\mu}_T = 7.15001$$

and

$$\hat{\sigma}_T = 1.06135$$

Example 4 Singly-Left Truncated

To insure meeting a lower weight specification limit of $TL = 0.1231$ in. on the thickness of a certain insulating washer, all production of this component is sorted through go, no-go gages, and all of thickness less than this value are discarded. For a random sample of 100 washers selected from the screened (i.e.

the retained) production, it is found that $\bar{x} = 0.124624$ and $s^2 = 2.1106 \cdot 10^{-6}$. Accordingly, $\gamma_T = \frac{s^2}{(\bar{x} - TL)^2} = \frac{2.1106 \cdot 10^{-6}}{(0.124624 - 0.12381)^2} = 0.9087$.

Cohen's MLE's: With $\gamma_T = 0.9087$ the maximum likelihood estimators of μ and σ cannot be computed by Cohen's method since Cohen's tables of the function λ_T are restricted for values of $\gamma_T \in [0.06246, 0.880]$.

MLELRT estimators: The maximum likelihood estimates obtained via these estimators are obtained by applying the computer program "*MLELRT.Single.Truncated*" with $\bar{x} = 0.124624$, $s^2 = .0000021106$, and $TL = 0.1231$ as shown in the Appendix. Accordingly, we obtain

$$\hat{\xi}_T = 3.72943 \quad \text{and} \quad \hat{\lambda}_T = 16.60941$$

$$\hat{\mu}_T = 0.09931$$

and

$$\hat{\sigma}_T = 0.00638$$

Example 5 Singly-Left Truncated and Singly-Left Censored Data

A simulated data set of size $n = 35$ was obtained from a truncated normal population with mean $\mu = 10$ and standard deviation $\sigma = 5$. The truncation limit is assumed to be the 10^{th} percentile of the population ($TL = 3.592242$). The left-truncated data are:

11.181363	11.358167	22.687924	7.421837	5.354584	22.844289	5.700818
12.443955	9.203403	6.388284	9.051160	14.314056	12.364398	8.104549
13.732982	8.201491	10.313142	14.009218	11.472960	8.480296	11.391664
17.913711	12.593001	6.555978	6.969704	8.462757	11.620034	3.951749
13.444815	15.265961	15.472041	9.237081	6.344069	9.826174	8.093112

The detection limit is assumed to be the 25^{th} percentile of the population ($DL = 6.62755$). Any value in the data set less than this detection limit was then taken as "below the detection limit or left censored" for the purpose

of applying the estimators introduced in Section 4. The left-truncated and left-censored data are:

11.181363	11.358167	22.687924	7.421837	< 6.627551	22.844289
< 6.627551	12.443955	9.203403	< 6.627551	9.051160	14.314056
12.364398	8.104549	13.732982	8.201491	10.313142	14.009218
11.472960	8.480296	11.391664	17.913711	12.593001	< 6.627551
6.969704	8.462757	11.620034	< 6.627551	13.444815	15.265961
15.472041	9.237081	< 6.627551	9.826174	8.093112	

There are 6 left censored data values, i.e., the censoring level in this data set is 17.1%. For these data, $n = 35$, $m = 29$ and $m_c = 6$. The sample mean and the sample standard deviation of the non-censored sample values are $\bar{x}_m = 11.9819$ and $s_m = 3.95799$, respectively. Applying the computer program “*MLELTLC.Single.LeftTrunc.LeftCens*” for these data, as shown in the Appendix, yields:

$$\hat{\xi}_T = -1.4756077 \quad \text{and} \quad \hat{\xi}_D = -0.3044654$$

Substitution into Equations (4.25) and (4.26) yields:

$$\hat{\lambda}_{\mu_{LTLC}} = 0.3814577 \quad \text{and} \quad \hat{\lambda}_{\sigma_{LTLC}} = 0.2881037$$

We employ estimators (4.23) and (4.24) to calculate the MLELTLC estimates of μ and σ as:

$$\begin{aligned} \hat{\sigma}_{LTLC} &= \sqrt{15.66568 + 0.2881037 \left(11.9819 - \frac{3.592242 + 6.62755}{2} \right)^2} \\ &= 5.41029 \end{aligned}$$

and

$$\hat{\mu}_{LTLC} = 11.9819 - 0.3814577(5.41029) = 9.918108$$

Inspection of the previous estimates reveals that MLELTLC estimators $\hat{\mu}_{LTLC}$ and $\hat{\sigma}_{LTLC}$ have the ability to recover the true mean and true standard deviation from singly-truncated and singly-left normally distributed data.

Example 6 Singly-Left Truncated and Singly-Right Censored Data

A simulated data set of size $n = 25$ was obtained from a truncated normal population with mean $\mu = 25$ and standard deviation $\sigma = 10$. The truncation limit is assumed to be the 25th percentile of the population ($TL = 18.2551$).

The left-truncated data are:

28.62091 23.06352 31.20407 20.19867 31.42485 **47.15421** 36.63650 **37.93222**
 24.80185 **42.67600** 34.32325 29.79365 36.66639 22.29325 **41.32932** 29.18277
 32.89069 **40.78423** 21.85124 31.23180 22.45349 21.73606 19.19825 18.70321
 19.98436

The detection limit is assumed to be the 80th percentile of the data set ($DL = 36.91956$). Any value in the data set greater than this detection limit was then taken as “above the detection limit or right censored” for the purpose of applying the estimators introduced in Section 5. The left-truncated and right-censored data are then given by

28.62091 23.06352 31.20407 20.19867 31.42485 > **36.91956** 36.63650
 > **36.91956** 24.80185 > **36.91956** 34.32325 29.79365 36.66639 22.29325
 > **36.91956** 29.18277 32.89069 > **36.91956** 21.85124 31.23180 22.45349
 21.73606 19.19825 18.70321 19.98436

There are 5 right-censored data values, i.e., the censoring level in this data set is 20.0%. For these data, $n = 25$, $m = 20$ and $m_c = 5$. The sample mean and the sample standard deviation of the non-censored sample values are $\bar{x}_m = 26.81294$ and $s_m = 5.860213$, respectively. Applying the computer program “*MLELTRC.Single.LeftTrunc.RightCens*” for these data, as shown in the Appendix, yields

$$\hat{\xi}_T = -1.3917255 \quad \text{and} \quad \hat{\xi}_D = 0.9102249$$

Substitution into Equations (5.7) and (5.8) yields:

$$\hat{\lambda}_{\mu_{LTRC}} = -0.1571824 \quad \text{and} \quad \hat{\lambda}_{\sigma_{LTRC}} = 92.00801$$

We employ estimators (5.5) and (5.6) to calculate the MLELTRC estimates of μ and σ as

$$\hat{\sigma}_{LTRC} = \sqrt{34.34209641 + 92.00801(26.81294 - \frac{18.2551 + 36.91956}{2})^2} = 9.461366$$

and

$$\hat{\mu}_{TD} = 26.81294 + (-0.15718247)(9.461366) = 25.32578$$

Inspection of the previous estimates reveals that MLELTRC estimators $\hat{\mu}_{LTRC}$ and $\hat{\sigma}_{LTRC}$ have the ability to recover the true mean and true standard deviation from singly-truncated and singly-right normally distributed data.

8. Conclusions and Recommendations

This paper has dealt with the problem of estimating the mean and standard deviation of a normal population from singly-left truncated data, and from both singly-left truncated and singly-left censored data. To calculate the maximum likelihood estimates of the mean and standard deviation of a normally distributed population from samples under consideration, it is necessary to solve simultaneously complex nonlinear estimating equations. Maximum likelihood estimating equations are derived which can be readily solved by simple iterative processes. New maximum likelihood estimators of population parameters from normally distributed left-truncated samples that contain left-censored data values are introduced. To facilitate the application of these estimators three computer programs are written in the R language: (1) “*MLET.Single.Truncated*” program is used to compute the MLELRT estimators from singly-left or singly-right truncated samples, (2) “*MLETD.Single.LeftTrunc.LeftCens*” program is used to compute the MLELTLC estimators from both singly-left truncated and singly-left censored samples, and (3) “*MLETD.Single.LeftTrunc.RightCens*” program is used to compute the MLELTRC estimators from both singly-left truncated and singly-

right censored samples. In addition, based on the new algorithm described in this article a computer program called “*Abou.Cohen.lambdat.hat*” to compute Cohen’s λ -parameter as a function of γ_T is written in both the S-PLUS and R languages. Copies of the source codes of the programs with some applications are given in the Appendix. When it is desired to estimate the mean and standard deviation of a normal distribution from truncated data, it is preferable to use the new algorithm method estimators described in this article. These estimators reduce the computation effort to determine the maximum likelihood estimates substantially by solving a nonlinear equation for the argument that is required to calculate the λ_T -parameter.

These estimators required no tables and are more easily computed than the MLE estimators of Cohen (1959). Cohen’s (1959) estimators cannot be used to obtain the maximum likelihood estimates from truncated samples that yield $\gamma_T < 0.06246$ or $\gamma_T > 0.880$. The new algorithm method for calculating the maximum likelihood estimators from truncated samples provides more adequate estimates for the population parameters and handles more situations than the existing Cohen’s (1959) estimators do.

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Appendix

Computer Programs

(1) Left or Right Singly-Truncated data:

The following computer program “MLET.Single.Truncated” has been written to compute the maximum likelihood estimators $\hat{\mu}_T$ and $\hat{\sigma}_T$ from both left and right singly-truncated samples utilizing the solution of (3.12). This computer program can be used for both S-PLUS and R. $\bar{x}$ and $ssquare$ are the sample mean and sample variance of the sample data set. TL is the truncation limit. $data$ is the data set entered as $data <- c(x_1, x_2, \dots, x_n)$. The output of this program is $\hat{\xi}_T = xithat$, $\hat{\lambda}_T = lambdat.hat$, $\hat{\mu}_T = mut.hat$ and $\hat{\sigma}_T = sigmat.hat$, respectively.

Case1: Given a data set

```
MLELRT.Single.Truncated<-function(data,TL) {
  n<-length(data)
  xbar<- mean(data)
  ssquare<-sum((data-xbar)^2)/n
  xit<-seq(-5,5,0.0001)
  N<-length(xit)
  gammat<-ssquare/(xbar-TL)^2
  odd.norm<-dnorm(xit)/(1-pnorm(xit))
  gxit<-(1-(odd.norm*(odd.norm-xit)))/(odd.norm-xit)^2 - gammat
  f<-gxit
  for(i in 2:N){
    if(f[i-1]<0 && f[i]>0) break
    xit.hat<-((xit[i]-xit[i-1])/(f[i]-f[i-1]))*(0.00-f[i-1]) + xit[i-1]
  }
  odd.normhat<-dnorm(xit.hat)/(1 - pnorm(xit.hat))
  lambdat.hat<-odd.normhat/(odd.normhat - xit.hat)
```

```

mut.hat<-xbar-lambdat.hat*(xbar-TL)
sigmat.hat<-sqrt(ssquare+lambdat.hat*(xbar-TL)^2)
list(xit.hat=xit.hat,lambdat.hat=lambdat.hat,mu.hat=mut.hat,
sigmat.hat=sigmat.hat)
}

```

Application: Data set given in Example 1

```

-----
data.example.1<-c
  (18.91943, 19.96156, 19.13235, 16.75358, 18.64818,
   12.42409, 21.02902, 11.14165, 19.44774, 15.14847, 22.42261,
   14.62704, 13.56455, 16.07526, 18.82385, 17.95926, 12.40669,
   10.84799, 14.77135, 12.88172)
> MLELRT.Single.Truncated(data.example.1, 10.79189)
$xit.hat [1] -1.080541
$lambdat.hat [1] 0.1931891
$mut.hat [1] 15.27568
$sigmat.hat [1] 4.149511

```

Case 2: Given summary statistics

```

MLELRT.Single.Truncated <-function(xbar,ssquare,TL) {
  xit<-seq(-5,5,0.0001)
  N<-length(xit)
  gammat<-ssquare/(xbar-TL)^2
  odd.norm<-dnorm(xit)/(1-pnorm(xit))
  gxit<-(1-(odd.norm*(odd.norm-xit)))/(odd.norm-xit)^2 - gammat
  f<-gxit
  for(i in 2:N){
    if(f[i-1]<0 && f[i]>0) break
  }
}

```

```

xit.hat<-((xit[i]-xit[i-1])/(f[i]-f[i-1]))*(0.00-f[i-1]) + xit[i-1]
}
odd.normhat<-dnorm(xit.hat)/(1 - pnorm(xit.hat))
lambdat.hat<-odd.normhat/(odd.normhat - xit.hat)
mut.hat<-xbar-lambdat.hat*(xbar-TL)
sigmat.hat<-sqrt(ssquare+lambdat.hat*(xbar-TL)^2)
list(xit.hat=xit.hat,lambdat.hat=lambdat.hat,mu.hat=mut.hat,
sigmat.hat=sigmat.hat)
}

```

Application: Example 2

```
> MLELRT.Single.Truncated(9.35,1.1264,12.00)
```

```
$xit.hat [1] -2.406196
```

```
$lambdat.hat [1] 0.009159304
```

```
$mut.hat [1] 9.374272
```

```
$sigmat.hat [1] 1.091202
```

Example 3

```
> MLELRT.Single.Truncated(7.15,1.1264,12)
```

```
$xit.hat: [1] -4.569548
```

```
$lambdat.hat: [1] 2.551735e-006
```

```
$mut.hat: [1] 7.150012
```

```
$sigmat.hat: [1] 1.061348
```

Example 4

```
> MLELRT.Single.Truncated(0.124624,.0000021106,0.1231)
```

```
$xit.hat: [1] 3.72943
```

\$lambdat.hat: [1] 16.60941

\$mut.hat: [1] 0.09931126

\$sigmat.hat: [1] 0.006378653

(2) Left-Singly Truncated and Left-Singly Censored data:

The following computer program “MLELTLC.Single.LeftTrunc.LeftCens” has been written to compute the maximum likelihood estimators $\hat{\mu}_{TD}$ and $\hat{\sigma}_{TD}$ from singly left-truncated and left-censored samples utilizing the solution of (4.20). This computer program can be used for both S-PLUS and R. $\bar{x}_m$ and $s_{msquare}$ are the sample mean and sample variance of the non-censored observations $x_1, x_2, \dots, x_m$. TL is the truncation limit and DL is the detection limit. The output of this program is $(\hat{\xi}_T, \hat{\xi}_D) = xiTD.hat$, $\hat{\lambda}_{\sigma_{LTLC}}, \hat{\mu}_{LTLC}$, $\hat{\lambda}_{\mu_{LTLC}} = sigmaTD.hat$ and $\hat{\mu}_{LTLC} = muTD.hat$, respectively.

```
MLELTLC.Single.LeftTrunc.LeftCens<-
```

```
function(n,mc,xmbar,smsquare,TL,DL){
```

```
  m<-n-mc
```

```
  xiT<-seq(-5,5,0.1)
```

```
  N<-length(xiT)
```

```
  hmc<-mc/n
```

```
  hm<-m/n
```

```
  h<-mc/m
```

```
  xiDf<-qnorm(hmc)
```

```
  xiTf<-(TL-xmbar)/sqrt(smsquare)
```

```
  dTD.hat<-xiDf-xiTf
```

```
  xiD<-xiT+dTD.hat
```

```
  TDLbar<-(TL+DL)/2
```

```

xiTDbar<-(xiT+xiD)/2
gammat<-((smsquare)/(xmbar-TDLbar)^2)
ZxiT<-dnorm(xiT)/(1-pnorm(xiT))
ZxiD<-dnorm(-xiD)/(1-pnorm(-xiD))
YxiTD<-(1/hm)*ZxiT + h*ZxiD-xiTDbar )
YxiT<-(ZxiT/hm)*((ZxiT/hm)-xiT)
YxiD<-(h*ZxiD)*((h*ZxiD)-xiD)
gxiT<-((smsquare)/(xmbar-TDLbar)^2)*YxiTD^2 + YxiT + YxiD +
2*(h/hm)*ZxiT*ZxiD - 1
g<-gxiT
for(i in 1:N) { #for1
fi<-is.na(g[i])
if(fi==T){
xhat<-NA
}
else { #else1
if(g[1]<0){
for(i in 2:N) {
if(g[i-1]<0 && g[i]>0) break
#xhat<-((xiT[i]-xiT[i-1])/(g[i]-g[i-1]))*(0.00-g[i-1]) + xiT[i-1]
xhat<-((xiT[i+1]-xiT[i])/(g[i+1]-g[i]))*(0.00-g[i]) + xiT[i]
}}
else{ ##else_aaa
for(i in 2:N) { #for2
if(!(g[i-1]>0) && !(g[i]<0)) {
xhat<-NA
}
else {
if(g[i-1]>0 && g[i]<0) break

```

```

#xhat<-((xiT[i]-xiT[i-1])/(g[i]-g[i-1]))*(0.00-g[i-1]) + xiT[i-1]
xhat<-((xiT[i]-xiT[i-1])/(g[i]-g[i-1]))*(0.00-g[i-1]) + xiT[i-1]
}
#if(g[i-1]<0 && g[i]>0) break
#xhat<-((xiT[i]-xiT[i+1])/(g[i+1]-g[i]))*(0.00-g[i]) + xiT[i]
} #for2
} ##else_aaa
} #else1
xiT.hat<-xhat
xiD.hat<-xiT.hat+dTD.hat
} #for1
xiTD.hat<-c(xiT.hat,xiD.hat)
xiTD.hat
xiTDbar.hat<-(xiTD.hat[1]+xiTD.hat[2])/2
ZxiT.hat<-dnorm(xiTD.hat[1])/(1-pnorm(xiTD.hat[1]))
ZxiD.hat<-dnorm(-xiTD.hat[2])/(1-pnorm(-xiTD.hat[2]))
YxiTD.hat<-(1/hm)*ZxiT.hat + h*ZxiD.hat-xiTDbar.hat )
YxiT.hat<-(ZxiT.hat/hm)*((ZxiT.hat/hm)-xiTD.hat[1])
YxiD.hat<-(h*ZxiD.hat)*((h*ZxiD.hat)-xiTD.hat[2])
lambdasigmaTD.hat<-
  (1/(YxiTD.hat)^2)*(YxiT.hat+YxiD.hat+2*(h/hm)*ZxiT.hat*
  ZxiD.hat)
lambdamuTD.hat<-(1/hm)*ZxiT.hat+h*ZxiD.hat
sigmaTD.hat<-sqrt(smsquare+lambdasigmaTD.hat*(xmbar-
  TDLbar)^2)
muTD.hat<-xmbar-lambdamuTD.hat*sigmaTD.hat
list(xiTD.hat=xiTD.hat, lambdasigmaLTLc.hat=lambdasigmaTD.hat,
lambdamuLTLc.hat=lambdamuTD.hat,

```

```
sigmaTLC.hat=sigmaTD.hat,muTLC.hat=muTD.hat)
}
```

Application: Simulated Data Set in Example 5

```
-----
> MLELTLT.Single.LeftTrunc.LeftCens(35,6,11.9819,15.66568484,
    3.592242,6.62755)
$xiTD.hat
[1] -1.4756058 -0.3044644
$lambdaTLC.hat
[1] 0.2881045
$lambdaTLC.muTLC.hat
[1] 0.3814581
$sigmaTLC.hat
[1] 5.410292
$muTLC.hat
[1] 9.9181
```

(3) Left-Singly Truncated and Right-Singly Censored data:

The following computer program “MLELTRC.Single.LeftTrunc.LeftCens” has been written to compute the maximum likelihood estimators $\hat{\mu}_{LTRC}$ and $\hat{\sigma}_{LTRC}$ from singly left-truncated and right-censored samples utilizing the solution of (5). This computer program is written in the R languages. $\bar{x}_m$ and s_m^2 are the sample mean and sample variance of the non-censored observations $x_1, x_2, \dots, x_m$. TL is the truncation limit and DL is the detection limit. The output of this program is $(\hat{\xi}_T, \hat{\xi}_D) = (xiTD.hat, \hat{\lambda}_{\sigma_{LTRC}}, \hat{\lambda}_{\mu_{LTRC}}, \hat{\sigma}_{LTRC}) = (sigmaTD.hat, muTD.hat)$, respectively.

```

MLELTRC.Single.LeftTrunc.RightCens<-
function(n,mc,xmbar,smsquare,TL,DL){
  xiT<-seq(-5,5,0.1)
  N<-length(xiT)
  m<-n-mc
  hmc<-mc/n
  hm<-m/n
  h<-mc/m
  sm<-sqrt(smsquare)
  xiDf<-qnorm(hm) #####      hm<-m/n
  xiTf<-(TL-xmbar)/sm
  dTD.hat<-xiDf-xiTf
  xiD<-xiT+dTD.hat
  TDLbar<-(TL+DL)/2
  xiTDbar<-(xiT+xiD)/2
  gammat<-((sm^2)/(xmbar-TDLbar)^2)
  ZxiT<-dnorm(xiT)/(1-pnorm(xiT))
  ZxiD<-dnorm(xiD)/(1-pnorm(xiD))
  YRxiTD<-((1/hm)*ZxiT - h*ZxiD-xiTDbar )
  YRxiT<-(ZxiT/hm)*((ZxiT/hm)-xiT)
  YRxiD<-(h*ZxiD)*((h*ZxiD)+xiD)
  gRxiT<-((sm^2)/(xmbar-TDLbar)^2)*YRxiTD^2 + YRxiT + YRxiD - 2*
  (h/hm)*ZxiT*ZxiD - 1
  g<-gRxiT
  gmin<-min(g)
  if(!any(g<0)){
    xhat<-xiT[g==gmin]
  }
}

```

```

}
else {
for(i in 1:N-1) {
if(g[i]>0 && g[i+1]<0) break
xhat<-((xiT[i+1]-xiT[i])/(g[i+1]-g[i]))*(0.00-g[i]) + xiT[i]
}}
xiT.hat<-xhat
xiD.hat<-xiT.hat+dTD.hat
xiTD.hat<-c(xiT.hat,xiD.hat)
xiTD.hat
xiTDbar.hat<-(xiTD.hat[1]+xiTD.hat[2])/2
ZxiT.hat<-dnorm(xiTD.hat[1])/(1-pnorm(xiTD.hat[1]))
ZxiD.hat<-dnorm(xiTD.hat[2])/(1-pnorm(xiTD.hat[2]))
YRxiTD.hat<-((1/hm)*ZxiT.hat - h*ZxiD.hat-xiTDbar.hat )
YRxiT.hat<-(ZxiT.hat/hm)*((ZxiT.hat/hm)-xiTD.hat[1])
YRxiD.hat<-(h*ZxiD.hat)*((h*ZxiD.hat)+xiTD.hat[2])
lambdaSigmaRTD.hat<-(1/(YRxiTD.hat)^2)*(YRxiT.hat+YRxiD.hat-2
*(h/hm)*ZxiT.hat*ZxiD.hat)
lambdamuRTD.hat<-(1/hm)*ZxiT.hat-h*ZxiD.hat
sigmaRTD.hat<-sqrt(sm^2+lambdaSigmaRTD.hat*
(xmbar-TDLbar)^2)
muRTD.hat<-xmbar+lambdamuRTD.hat*sigmaRTD.hat
list(xiTd.hat=xiTD.hat,lambdaSigmaLTRC.hat=lambdaSigmaRTD.hat,
lambdamuLTRC.hat=lambdamuRTD.hat,
sigmaLTRC.hat=sigmaRTD.hat,muLTRC.hat=muRTD.hat)
}

```

Application: Simulated Data Set in Example 6

```
-----
> MLELTRC.Single.LeftTrunc.RightCens
      (25,5,26.81294,34.34209641, 18.2551,36.91956)

$xiTd.hat
[1] -1.3917255 0.9102249
$lambdaSigmaLTRC.hat
[1] 92.00801 $lambdamuLTRC.hat
[1] -0.1571824 $sigmaLTRC.hat
[1] 9.461366
$muLTRC.hat
[1] 25.32578
```

(4) Computing Cohen's λ -parameter for Singly-Truncated data:

The following computer program "Abou.Cohen.lambdat.hat" has been written to compute the Cohen λ -parameter λ_T corresponding to any value of $\gamma_T = \frac{s^2}{(\bar{x} - TL)^2}$ tabulated in Cohen (1959). In addition, this program can be used to compute the λ -parameter given any value of γ_T falls in or outside the interval $[0.06246, 0.880]$.

```
Abou.Cohen.lambdat.hat<-function(gammat) { #####cccc
  solvegt<-function(gammat){ ##bb
  myfun<-function(xit,gammat) { ###aa
  odd.norm<-dnorm(xit)/(1-pnorm(xit))
  (1-(odd.norm*(odd.norm-xit)))/(odd.norm-xit)^2 - gammat
  } ###a
  outhat<-uniroot(myfun,lower=-5.0,upper=5.0,tol=0.0001,
    gammat=gammat)
```

```

xithat<-outhat$root
xithat
} ##bb
xit.hat<-solvegt(gammat)
odd.normhat <- dnorm(xit.hat)/(1 - pnorm(xit.hat))
lambdat.hat<-odd.normhat/(odd.normhat - xit.hat)
list(xit.hat=xit.hat, lambdat.hat=lambdat.hat)
} ####cccc
Application: gammat = 0.62991
-----
> Abou.Cohen.lambdat.hat(0.62991)
$xit.hat: [1] 0.3499823
$lambdat.hat: [1] 1.512237

```